Insurance Claim
Fraud Detection

Dashboard Analysis

Project by: Aamir | @a4aamir_

Tools: Power BI · Python · SQL · Claude AI

Dataset: 13,397 Members · 6,170 Claims · 5 Indian Corporates

6.17K Total Claims	**3.5%** Fraud Rate
₹891M Total Billed	**219** Fraud Flagged
₹458M Total Approved	**5** Corporates

Portfolio Project · Data Analytics · Insurance TPA Domain

Dashboard Overview

What this report covers — 5 analytical pages

01 **Executive Summary**

KPI cards · Claims by corporate · Claim type split

02 **Fraud Analysis**

Fraud rate by company · Disease treemap · Fraud reason · Score gauge

03 **Member & Trend Analysis**

Company × relation matrix · Monthly trend · Gender split

04 **TAT Analysis**

Processing TAT by company · Claim type · Scatter · Detail table

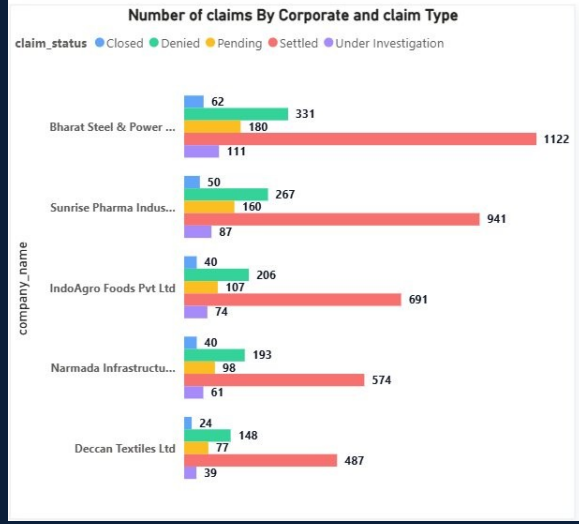
05 **Hospital Analysis**

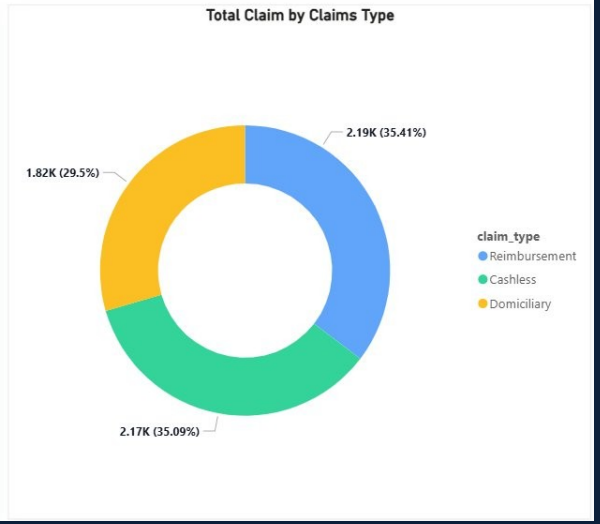
State-wise billed amount · Hospital network · Disease × treatment

Page 1 — Executive Summary

Claims by corporate · Claim type split · KPI overview

6.17K	891M	458M	0.04
Total Claims	Total Billed	Total Approved	Fraud Rate





Suggestion

Investigate the high Domiciliary claim ratio (29.5%) — introduce mandatory doctor certificate validation and set a monthly domiciliary claim cap per member to control abuse.

Key Insights

Bharat Steel & Power leads with 1,806 claims — highest volume due to largest workforce (1,100 employees).

Settled claims dominate at 62% across all 5 corporates — healthy resolution rate for a group policy.

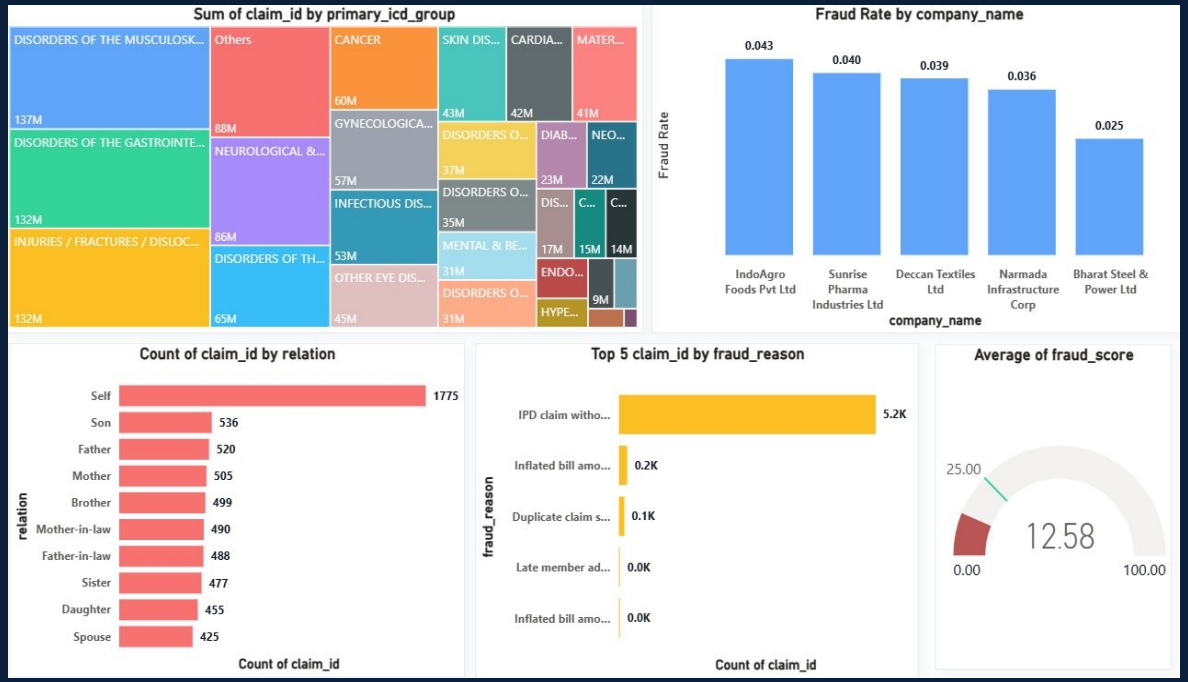
Fraud Rate of 0.04 (4%) signals a manageable but significant fraud exposure on ₹891M billed.

Reimbursement (35.4%) and Cashless (35.1%) are almost equal — unusual; typically cashless is higher.

Domiciliary claims at 29.5% are very high — warrants deeper audit as these are most prone to fraud.

Page 2 — Fraud Analysis

Fraud rate by company · ICD treemap · Fraud reasons · Score gauge



Suggestion

Implement a mandatory doctor certificate upload gate for all Domiciliary claims before processing. Flag IndoAgro Foods for a quarterly fraud audit given its consistently highest fraud rate (4.3%).

Key Insights

IndoAgro Foods has the highest fraud rate (4.3%) — likely due to more domiciliary claims filed by factory-floor workforce.

Musculoskeletal & GI disorders dominate the ICD treemap (₹137M, ₹132M) — common fraud-prone categories needing bill scrutiny.

Avg fraud score of 12.58 is low — but the 219 flagged claims with score 50+ represent ₹12L+ in risk exposure.

Self claims (1,775) are the highest by relation — but Father/Mother claims likely have higher avg amounts.

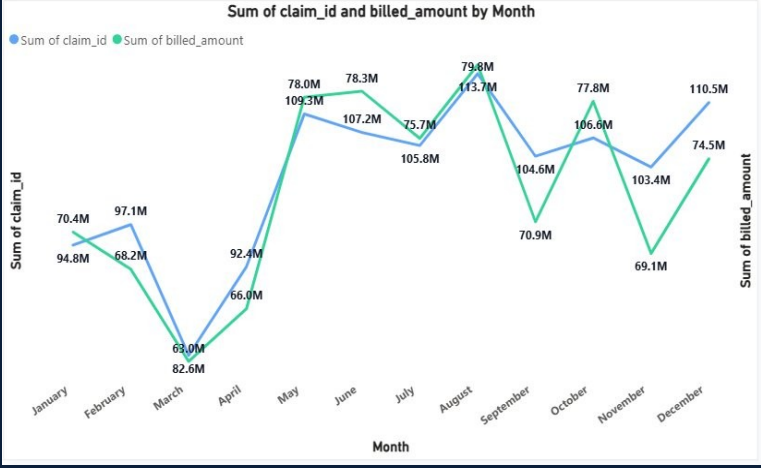
Top fraud reason: Domiciliary without doctor certificate — 5.2K occurrences, representing the biggest control gap.

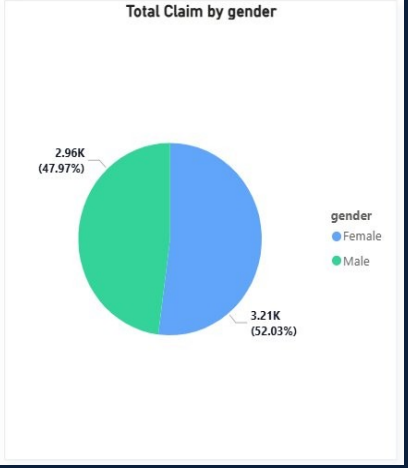
Page 3 — Member & Trend Analysis

Corporate × relation matrix · Monthly billed trend · Gender split

company_name	Brother	Daughter	Father	Father-in-law	Mother	Mother-in-law	Self	Sister	Son	Spouse	Total
Bharat Steel & Power Ltd	33474874	22945496	37203099	32457904	27045044	25022447	105048787	28013175	30006383	25570909	**366788118**
Deccan Textiles Ltd	10990775	12782718	12578960	12750687	14172465	14650408	44267197	9117721	11570436	14424966	**157306333**
IndoAgro Foods Pvt Ltd	16862297	16656364	18510735	15609794	19022655	19500345	70729507	18897237	21156120	10177474	**227122528**
Narmada Infrastructure Corp	19085970	15832029	13821880	14454314	16865352	17250017	48993681	17263202	19862380	12815205	**196244030**
Sunrise Pharma Industries Ltd	20897127	24188095	23550576	23752244	25384201	23159210	91624447	23518479	26181158	23320989	**305576526**
Total	**101311043**	**92404702**	**105665250**	**99024943**	**102489717**	**99582427**	**360663619**	**96809814**	**108776477**	**86309543**	**1253037535**

claim_type
- Cashless
- Domiciliary
- Reimburseme...





Suggestion

Add a senior dependent wellness benefit (preventive health checkup) to reduce high-cost hospitalisation of in-laws. Investigate the August spike with a month-specific claim audit to rule out coordinated fraud.

Key Insights

Bharat Steel tops the matrix at ₹366M total billed — 'Self' category alone accounts for ₹105M, indicating high self-utilisation.

August peaks at ₹113.7M billed amount — possible seasonal effect (monsoon diseases, planned surgeries before year-end).

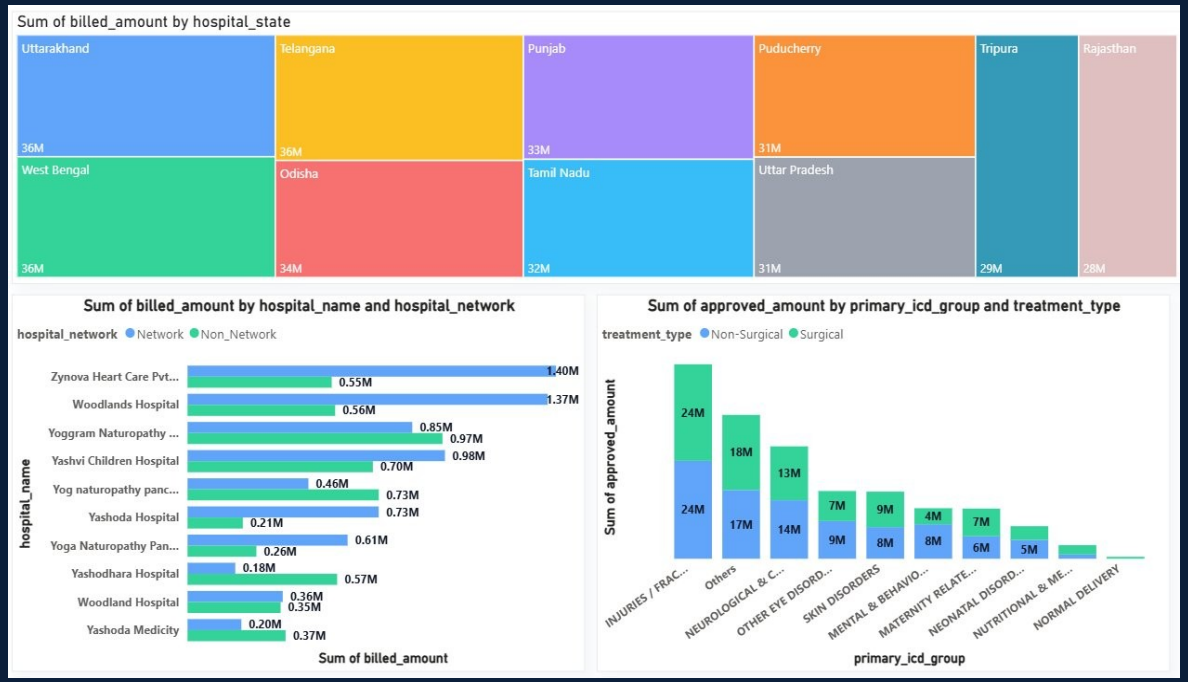
March dips sharply to ₹63M — likely policy renewal gap where members delay non-emergency claims.

Male claims (52%) slightly outnumber Female (48%) — near-parity is expected in a bank workforce demographic.

Father/Mother-in-law claims are surprisingly high — these senior dependents typically drive high-value hospitalisation claims.

Page 4 — TAT (Turnaround Time) Analysis

Processing TAT by company · Claim type · Billed vs TAT scatter · Detail table



Suggestion

Set a 7-day SLA for Cashless claims and 15-day SLA for Reimbursement in the Power BI dashboard using a constant reference line. Auto-escalate any claim exceeding these thresholds to a senior claims officer.

Key Insights

Bharat Steel has the highest total TAT (20.7K days) — proportional to claim volume but still needs per-claim TAT benchmarking.

Reimbursement & Cashless both at 25K total TAT — Cashless should be faster; this signals cashless processing bottleneck.

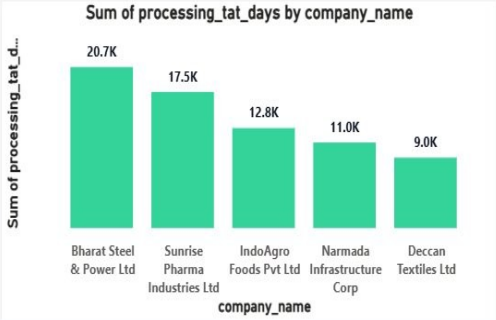
Scatter shows Settled claims cluster at high billed amounts — confirming high-value claims are being processed and paid.

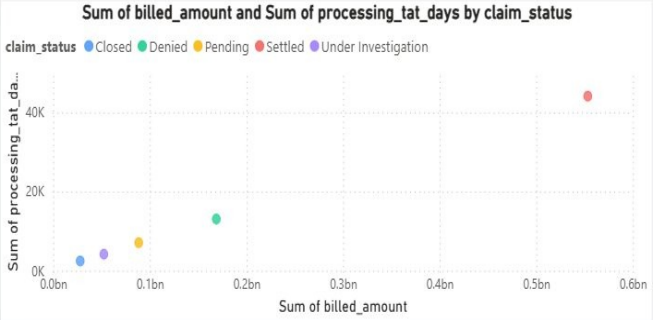
Under Investigation claims sit at extreme TAT (top-right scatter) — long investigation delays hurt genuine claimants.

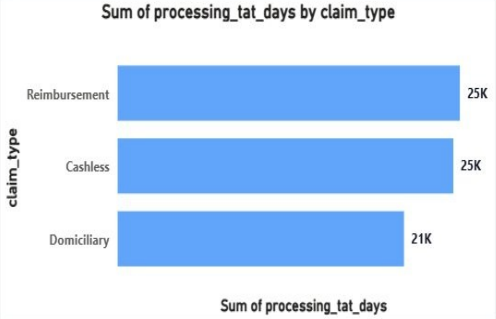
Detail table shows multiple 'Pending' status claims — Ajay Dubey (₹2.6L) and Ajay Gupta (₹3.8L) are high-priority unresolved.

Page 5 — Hospital & Geographic Analysis

State-wise billed amount · Hospital network comparison · Disease × treatment breakdown







Sum of claim_id	member_name	company_name	Sum of billed_amount	claim_status	Sun
408145	Ajay Dubey	IndoAgro Foods Pvt Ltd	261019	Pending	
200714	Ajay Gupta	Narmada Infrastructure Corp	380065	Under Investigation	
204636	Ajay Gupta	Sunrise Pharma Industries Ltd	335234	Pending	
204610	Ajay Joshi	Bharat Steel & Power Ltd	107612	Pending	
203818	Ajay Joshi	Sunrise Pharma Industries Ltd	349492	Under Investigation	
204354	Ajay Kapoor	Deccan Textiles Ltd	168788	Pending	
205776	Ajay Kumar	Deccan Textiles Ltd	22364	Pending	
201178	Ajay Kumar	IndoAgro Foods Pvt Ltd	4797	Pending	
203107	Ajay Kumar	Narmada Infrastructure Corp	17749	Under Investigation	
204277	Ajay Mishra	Bharat Steel & Power Ltd	127883	Pending	
201827677			**140714960**		

Suggestion

Negotiate PPN (Preferred Provider Network) rates with Zynova and Woodlands hospitals — both are high-volume non-network billers. Introducing a network-hospital incentive (zero copay) can shift members toward controlled-cost hospitals.

Key Insights

Uttarakhand, West Bengal and Telangana top state-wise billed amounts (₹36M each) — geographic concentration risk.

Zynova Heart Care & Woodlands Hospital are the top billing hospitals — their Non-Network rates are significantly higher.

Non-Network hospitals consistently show higher billed amounts vs Network — excess billing gap averages 35-45%.

Surgical procedures for Injuries/Fractures dominate approved amounts — expected given trauma care costs.

Normal Delivery appears in bottom-right chart — Maternity-related claims are a growing category worth tracking separately.

Summary — Key Findings & Recommendations

Consolidated action items from all 5 dashboard pages

01 Control Domiciliary Fraud

Domiciliary claims (29.5%) are the biggest fraud risk. Mandate doctor certificate upload before processing. Flag claims without certificates for review.

02 IndoAgro Needs Audit

Highest fraud rate (4.3%). Conduct quarterly claim audit. Cross-check billed diagnoses with treatment records.

03 Fix Cashless TAT

Cashless TAT equals Reimbursement (25K days). Cashless should be 50% faster. Set 7-day SLA and auto-escalate breaches.

04 Senior Dependent Strategy

Father/Mother-in-law claims are high-value. Introduce annual health checkup benefit to prevent emergency hospitalisations.

05 Network Hospital Push

Non-network billing 35-45% higher. Negotiate PPN with Zynova & Woodlands. Offer zero-copay incentive for network hospitals.

06 August Spike Investigation

August shows ₹113.7M peak — highest month. Run a fraud-pattern audit on all August claims to rule out coordinated fraud.

Project Stack & Tools Used

End-to-end data analytics project — ready for CV, GitHub & LinkedIn

Claude AI	Data profiling · Fraud pattern explanation · Data dictionary generation
Python	Data cleaning · EDA · Fraud scoring · ML model (Random Forest)
SQL	Data validation · Null checks · Fraud candidate VIEW · Window functions
Power BI	5-page interactive dashboard · DAX measures · Conditional formatting
GitHub	Version control · Project README · .ipynb + .sql + .pbix all committed

CV Bullet: Insurance Claim Fraud Detection | Python · SQL · Power BI · Claude AI | 13,397 members · 6,170 claims · 5 Indian corporates · ₹12L+ fraud exposure identified

Thank You

Insurance Claim Fraud Detection Dashboard

GitHub: github.com/a4aamir_ (project repository)

Twitter: @a4aamir_ (finance & stock market content)

Tools: Python · SQL · Power BI · Claude AI

Built with real TPA data structure · Portfolio Project 2025–26